Fiscal Year:	FY 2017	**Task Last Updated:**	FY 06/23/2017
PI Name:	Wu, Honglu Ph.D.		
Project Title:	Dependence of radiation quality on charged particle-induced early and late damages in chromosomes		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77058-3607	**Congressional District:**	22
Comments:	PI email changed from honglu.wu-1@nasa.gov after PI retirement from NASA.		
Project Type:	Ground	**Solicitation / Funding Source:**	2010 Space Radiobiology NNJ10ZSA001N
Start Date:	07/01/2011	**End Date:**	02/28/2017
No. of Post Docs:	2	**No. of PhD Degrees:**	2
No. of PhD Candidates:	2	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Simonsen, Lisa	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 2/28/2017 per S. Monk/LaRC (Ed., 9/9/16) NOTE: End date changed to 12/31/2016 per L. Simonsen/LaRC (Ed., 12/31/15) NOTE: End date changed to 12/31/2015 per PI (Ed., 11/13/14)		
Key Personnel Changes/Previous PI:	None		
COI Name (Institution):	Zhang, Ye (NASA Johnson Space Center)		
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Tumor formation in humans or animals is a multi-step process. An early stage of cancer development is believed to be genomic instability (GI), which accelerates the mutation rate in the descendants of the cells surviving radiation exposure. A relationship between GI and tumor metastasis, as well as between GI and adaptation to drug treatment, has also been suggested. Evidence for GI induced from radiation exposure has been established in the last two decades. GI is defined as elevated or persistent genetic damages occurring many generations after the cells are exposed, and can be manifested in several biological consequences including mutation, apoptosis and damages in chromosomes. Genomic instability can be induced by both low- and high-LET (linear energy transfer) radiation. Space radiation consists of a mixture of different types of ions covering a wide range of masses and energies. Its associated health risks to astronauts in flight are of special concern for prolonged space missions beyond the protective terrestrial magnetosphere. Before such missions can be undertaken, a much more detailed understanding of these biological effects is required to allow planning and implementation of protective countermeasures. The proposed research will attempt to investigate GI induced from exposures to energetic ions of the types and energies encountered during spaceflight, and will address the critical question of carcinogenesis risk and its dependence on radiation quality, which is an emphasized area of the current NASA Research Announcement (NRA). The overall goals of this proposed project are (1) to determine the relationship between the quality and fluence of energetic charged particles and the induction of delayed damages in chromosomes, which is commonly referred to as chromosomal instability and (2) to understand the relationship between chromosomal instability and other events at the cellular and molecular level that play a role in the maintenance of genome integrity. For this proposal, we select well-developed and widely used methodologies for quantification of GI. The three models include human T-lymphocytes collected from healthy subjects, and haemopoietic cells extracted from CBA/CaH and from C57BL/6J mice, which differ in radiosensitivity. The cells will be exposed in vitro to protons, Si, and Fe ions which represent three of the most important particle types in space. Chromosomal instability will be analyzed in both the total population of the cells surviving after multiple generations post irradiation (non-clonal assay), as well as in clonal populations derived from single cells surviving particle irradiation (clonal assay). The non-clonal assay is less labor intensive and allows analysis of samples exposed to particles of different types, energies, and doses, whereas the clonal assay will be designed to tackle the question of genetic background influence on the induction of GI. Several techniques for chromosome analysis will be used which include multi-color fluorescence in situ hybridization (mFISH) and the multi-color banding in situ hybridization (mBAND). Measurements of telomere length and expressions of genes involved in genomic stability will also be performed in cells from unstable, as well as stable, clones. The clonal expansion assay is arguably more suitable than the non-clonal assay for determining the correlation between chromosomal instability and other biological consequences. Key to our proposal is that the investigators have substantial expertise and published literature to ensure the success of the proposed studies. The Principal Investigator has extensive experience in the induction of heavy ion-induced chromosomal aberrations in human lymphocytes using mFISH, mBAND, and other modern chromosome painting techniques. The Co-Investigator was trained in immunology and has substantial expertise in T-lymphocyte isolation and expansion. Dr. Kadhim is a recognized expert in GI and had made significant contributions to the understanding of radiation-induced GI.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The research addresses the question of genomic instability from charged particle exposures in space. By assessing genomic instability quantitatively for different charges and energies of the particles, we aim at contributing to the reduction of uncertainties in the assessment of cancer risks in astronauts after long-duration spaceflights.
Task Progress:	To achieve the goals of the project, human and mouse cells were exposed to low- and high-LET (linear energy transfer) radiation for analysis of different endpoints such as chromosome aberrations and apoptosis. Initial radiation damage to the cells was assessed shortly after irradiation, and late damage was assessed after the cells were cultured for a period of weeks. The project specifically addresses the amount of the initial damage that will persist through multiple cell divisions, and whether greater levels of damage may be found after multiple cell divisions. The biological samples investigated in the project were: • Human lymphocytes isolated from test subjects • Human mammary epithelial cells (CH184B5F5/M10) • Mouse bone marrow cells extracted from CBA/CaH mice • Mouse bone marrow cells extracted from C57BL/6J mice • Immortalized human bronchial epithelial cells (HEBC3KT) • BALB/c and C57BL/6 mice The radiation types used in the project were protons, Si ions, and Fe ions generated at the NASA Space Radiation Laboratory (NSRL). The proton source at Loma Linda University was also used. A gamma source at Johnson Space Center (JSC) was used as low-LET controls. In this project, GI was investigated for both low- and high-LET radiation across different species and different cell types using both cell models and animal models. The degree of GI is apparently dependent on the quality of radiation, as well as on the biological systems. Consequently, the RBE (relative biological effectiveness) value can be different for different cell types and different endpoints. In our study, the RBE value for Fe ions can be high for initial chromosome aberrations in human lymphocytes. However, the value was found to be close to 1 when late damage was assessed. Early damage in the biological system does not necessarily predict residual damage, which is more closely related to late radiation risks such as cancer.
Bibliography Type:	Description: (Last Updated: 07/15/2026)

Articles in Peer-reviewed Journals	Lu T, Zhang Y, Wong M, Feiveson A, Gaza R, Stoffle N, Wang H, Wilson B, Rohde L, Stodieck L, Karouia F, Wu H. "Detection of DNA damage by space radiation in human fibroblasts flown on the International Space Station." Life Sci Space Res (Amst). 2017 Feb;12:24-31. Epub 2016 Dec 24. https://doi.org/10.1016/j.lssr.2016.12.004 ; PubMed PMID: 28212705 , Feb-2017
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